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	In space, astronauts are exposed to a complex and unique radiation environment composed of trapped proton and electron radiation, galactic cosmic rays (GCR; energetic protons and heavy charged ions, HZE), and solar particle events (SPE), primarily protons. Although these radiations can result in appreciable cumulative doses to astronauts during long-duration missions, the exposures occur at low fluences and fluence rates, such that particle traversals through individual cells in an astronaut's body may be well separated in tissue location and time. Under such conditions, so-called non-targeted effects such as bystander responses may have increased significance. The radiation-induced bystander effect is the occurrence of biological responses in unirradiated cells near to or sharing medium with cells actually traversed by radiation. We hypothesize that at the low fluences of protons and HZE particles to which astronauts are exposed in space, a significant fraction of the cellular changes may be due to bystander responses, in addition to the damages caused in cells directly traversed by ionizing particles or their associated delta rays. We have the		

Task Description:	following specific aims. 1. Test the hypothesis that bystander effects occur at particle fluences so low that only 1-2 % of cells in a culture dish are traversed by an energetic proton or HZE particle. 2. Extend our low fluence work to studies using repeated exposures of cells to low fluences and mixed (actually sequential) beams of more than one particle type. This will test the hypothesis that cells that have survived an initial traversal and sent a bystander signal, can again send a signal when traversed once again by a particle or can subsequently respond as bystanders to a hit in a neighboring cell. 3. Our studies have indicated that reactive oxygen species (ROS) are at least partially involved in transmittal of the Fe ion-induced bystander signal from hit to unirradiated cells. We will test the hypothesis that non-nuclear damage-initiated, metabolically produced ROS in hit cells are involved in the early steps of bystander signaling. Additionally, we will investigate chemical countermeasures to reduce the bystander responses based on removal of ROS or prevention of their formation. 4. In tissues, cells do not exist in isolation, and there is increasing interest in how signaling between cell types (especially stromal and epithelial) influences cellular responses to stresses. We will test the hypothesis that radiation-induced bystander signals can be sent between differing cell types, but the characteristics of the responses depend on the cell types and the types of radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although our studies are specifically geared to providing information on the possible biological effects, especially cancer induction, of low fluences of energetic particles on astronauts, the data may have relevance to cancer therapy and the effects of low doses of radiation on radiation workers (e.g., those involved in clean-up of radioactive sites) and the general public. With regards to cancer, there is increasing use of protons (energies of 65-250 MeV) as an effective treatment modality in radiation oncology because they provide a great therapeutic advantage in some types of cancers due to exquisite dose localization. Over 40,000 patients have been treated world-wide with protons, and several new proton centers have opened in the last few years or are under construction in the US. However, despite the clinical success of protons, some concerns exist regarding the possible side-effects, including the possibility of cancer caused by the low doses received by normal tissues. Additionally, there is increasing interest in the use of carbon ions in cancer therapy, with carbon ion treatment centers now in use in Japan and Germany, and the potential being discussed in the US. Our studies could provide important and useful new insight on the effects of these energetic particles on biological systems and the mechanisms by which they act. Such findings would be expected to translate into helping to develop effective ways to reduce possible side-effects, including cancer, from the treatments.
	Bystander effects induced at very low particle fluences (aim 1). Previously, we have reported using a transwell insert culture method to demonstrate media-mediated bystander responses in AG01522 normal human fibroblasts after X-irradiation, 1 GeV/n iron ions, and 1 GeV protons. The ion work was performed at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory. Although at doses above about 10 cGy the magnitude of the bystander effects is small compared to the damage from direct irradiation, at lower doses the bystander responses make a significant contribution to the overall cell damage measured. More recently, we have reported on our accumulating data from work at NSRL on extending studies to very low fluences of high energy iron ions and protons. We now have data from at least three independent experiments at low fluences of both iron ions and protons for the two endpoints of 53BP1 foci formation 5 h after irradiation (indicative of level of DNA double strand breaks) and micronuclei (MN) formation 72 h after irradiation (surrogate for chromosome aberrations). With iron ions, at fluences where 1% or less of the cells are traversed by an ion, the level of cells with MN or 53BP1 foci is the same as in unirradiated cells (4-5%). However, over a fluence range from about 2e3 to about 4.5e4 ions/cm2 (2-50% of cells traversed by an ion), the level of damage is 1.5-2-fold higher than in control cells, regardless of whether the cells are irradiated or bystander cells. Above about 50% of the cells traversed, the fraction of cells with damage increases rapidly with dose in the irradiated population, but in the bystander population there is no further increase in fraction of damaged cells. With protons, the same level of DNA damage occurs as with Fe ions, but the responses are shifted to higher fluences, i.e., the lowest fluence at which damage reaches the 50% increase level is 2e5 protons/cm2 (about 2 protons/cell) and the plateau continues to about 2e7 protons/cm2 (200 protons/cell), above which there is a rapid increase in damage in the irradiated population, but a continued plateau in the bystander cells. Based on these data, we hypothesize that: (1) in the low fluence plateau range where irradiated and bystander cell populations show the same level of damage, the effects measured in the irradiated population actually result from bystander (i.e., non-ion-traversed) cells, rather than from cells traversed by particles and (2) the range of fluence over which the plateau occurs depends on particle LET, although the level of cells expressing damage is independent of LET (and dose). These data clearly indicate the importance of investigating cellular responses at very low particle fluences. Bystander responses in mixed radiation fields (aim 2). As mentioned in previous annual reports, we have been among the pioneering groups at NSRL to use the mixed, actually sequential, beams of Fe ions and protons. These studies were designed to assess possible interactions between the two types of radiation in cells, when the irradiations were separated in time. Such studies are highly relevant to the space environment since it has been estimated that each cell in an astronaut's body would be traversed by a proton once every three days and by an HZE particle every few months. We presented preliminary data last year, but have more extensive data to report this year. These studies were conducted using 1 cGy doses of protons and Fe ions, which correspond to about 350 proton traversals and 0.45 Fe ion traversals per cell, under the conditions used. The data show that sequential exposures to 1 GeV protons and 1 GeV/n Fe ions cause the same magnitude of DNA damage (% cells with MN) in irradiated cells as seen with either protons or Fe ions alone. Furthermore, the same amount of DNA damage is also observed in non-irradiated bystander cells sharing medium with cells irradiated with either protons or iron ions or protons plus iron ions. However, when "bystander" cells are exposed to protons up to 3 h before co-culture with Fe ion-irradiated cells, no DNA damage in the "bystander" cells is observed. Persistent production of reactive oxygen species (ROS) has been shown in mammalian cells after exposure to ionizing radiation, and several studies have reported a role for ROS in bystander signaling. We investigated whether ROS could mimic the effect of prior proton irradiation in the reduction of the medium-mediated Fe-induced bystander response just described. Bystander cells were treated with a low level of H2O2 30 min, 3 h or 24 h before being put into co-culture

Task Progress:	with cells irradiated with Fe ions. Treatment with 5 microM H2O2 alone did not increase the frequency of MN formation. Furthermore, pretreatment of bystander cells with a low level of H2O2 did not modulate their ability to show increased MN formation after sharing medium with Fe-irradiated cells. The data imply that ROS may not be involved in the decrease in Fe ion-induced medium-mediated bystander effects caused by a low prior proton dose. Role of ROS in bystander signaling (aim 3). We have published data showing that addition of SOD, catalase or c-PTIO to medium in co-cultures of Fe-irradiated and unirradiated cells decreases the production of MN or ?-H2AX foci in bystander cells, supporting roles for superoxide, H2O2 and NO in bystander responses after doses of 0.1 Gy and higher. Interestingly, and unexpectedly, we have obtained preliminary data that suggest that at low fluences of Fe ions (<5e4 ions/cm2), catalase, SOD and c-PTIO do not decrease the bystander response. These studies are currently being repeated, but if they hold up they suggest that different mechanisms of intercellular signaling may be involved at high and low particle fluences. Bystander signaling in mixed cell types and 3D (aim 4). Previously, we had obtained preliminary data on signaling between mixed cell types, using AG01522 human fibroblasts and hTERT-immortalized human keratinocytes. The results suggested that with 250 kVp X-rays, and using the transwell insert system, bystander signaling occurs from irradiated fibroblasts to unirradiated keratinocytes and vice versa. However, with 1 GeV/n iron ions, our preliminary data suggested that the bystander response in keratinocytes co-cultured with irradiated fibroblasts is delayed, and no response is seen in the reverse direction. In more recent studies, we have confirmed the previous findings with 1 GeV/n iron ions and extended them to 1 GeV protons. The data show that there is a 1.5-2.0-fold increase in MN formation in bystander fibroblasts or keratinocytes after proton irradiation of either cell type. However, no significant increase in MN is seen in bystander fibroblasts in co-culture with irradiated keratinocytes. Data also indicate that a 1.5-2-fold increase in the fraction of cells expressing 53BP1 foci occurs in both cell types when either cell type is irradiated. In bystander cells, the foci occur earlier after proton irradiation (peak about 12 h) than after iron irradiation (peak between 24- 48 h). Thus, these data support our hypothesis that the occurrence of bystander responses depends on cell type, radiation quality and endpoint. Studies with 290 MeV/n carbon ions are in progress to further test for an LET dependence of the lack of signaling from keratinocytes to fibroblasts after iron ions. If LET is important, we predict that carbon ions, of intermediate LET, will give an intermediate response between that of protons and iron ions. This is an important study, as it may shed light on conditions for radiation quality dependence of the bystander effect. We have also developed a 3D construct model using the AG01522 fibroblasts and the h-TERT-immortalized keratinocytes. The fibroblasts are embedded in a collagen gel in a well of a 6-well plate, then the keratinocytes are layered on top of the gel, where they grow at the air-liquid interface to form a stratified, epithelial-like layer. X-ray dose response data show that the % fibroblasts or keratinocytes positive for 53BP1 foci in the 3D constructs increases with dose, although more foci are seen in the keratinocytes than in the fibroblasts. With either cell type, the fraction of cells with DNA damage in the 3D constructs is less than in cells irradiated in 2D monolayers (e.g., 1 h after 0.1 and 2 Gy to AG01522 monolayers about 35 and 95% of cells, respectively, are positive for 53BP1 foci). Hence, the data support the hypothesis that radiation response depends on cell organization in 2D versus 3D. Several years ago, the team at Columbia University showed in very elegant experiments using their particle beam to irradiate MatTek skin constructs, that micronuclei and apoptosis occur in unirradiated (bystander) cells at distances up to 1 mm from the irradiated cells. At NSRL, we have used an alternative approach of cutting a construct into two pieces, irradiating half of it, then immediately adding back the unirradiated (bystander) half into the insert containing the irradiated construct half. At subsequent times, both halves are fixed, sectioned and analyzed for effects. Preliminary data from the single experiment done to date show that 53BP1 foci could be quantified in fibroblasts of both the irradiated and bystander 3D construct halves at varying times after irradiation with 0.1 Gy of 1 GeV protons. In the irradiated half, substantial DNA damage is seen 1 h after irradiation, then decreases with time. In contrast, the level of damage in the bystander half is low at 1 h, but increases out to 48 h. These are very preliminary data, but the results demonstrate that the approach will work as a means to study bystander effects in a 3D model at NSRL, and we will pursue this in the coming year. More importantly from a scientific perspective, the data show that bystander signals can travel considerable distances (on the order of mm), consistent with the work of Belykov et al.
Bibliography Type:	Description: (Last Updated: 11/13/2018)
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